

MovieLOD

DBpedia reuse · OWL reasoning · linked movie knowledge

From movie data to a knowledge graph

Ontology · RDF · Linked Open Data · SPARQL

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30

real films

37

named classes

1,727

external links

Objectives and synchronized evidence

Ontology, data, publication, linking and SPARQL access

ID	Requirement	Deliverable / evidence
R1	DBpedia-based ontology	37 named classes; HermiT consistency
R2	Source-supported data	76 retained responses and provenance
R3	RDF publication	Canonical RDF / OWL 3.0.0
R4	Identity alignment	1,699 Wikidata + 28 DBpedia links
R5	SPARQL access	27 questions; source / inferred / named graph modes

Report: 15 pages · main slides: 24 · MP4 excluded by request. Examples: Inception and Nolan.

Architecture and knowledge layers

One reproducible pipeline across every interface

01

SOURCE FACTS

76 retained responses
+ provenance

02

BUILD 3.0.0

Canonical RDF / OWL
+ resource pages

03

REASON / CHECK

HermiT + OWL RL
27 query checks

04

QUERY MODES

Source / inference
Before / after

NEW MODEL: asserted.ttl + schema.ttl + inferred.ttl; query 27 uses before_after.trig.
APPLICATION: Flask / RDFLib / Comunica share the canonical version-3 graph exports.

Dataset: units and counting scope

Final OWL: 19,025 triples including schema and asserted facts

30

films

851

people

1,010

credit records

45

companies

672

award entities

76

source responses

37 named classes · 19 object + 5 datatype properties · 4 controlled role individuals

Ontology inventory: reuse before extension

17 DBpedia classes + 1 Void class + 19 local classes

17 classes	DBpedia classes and hierarchy
1 class	void:Dataset
3 classes	Contribution / Role / SourceSnapshot
2 classes	ActionGenre / DramaGenre
4 classes	Role-defined credit subclasses
10 classes	Domain inferred subsets

37 named classes: 17 dbo: + 1 void: + 19 ex:
10 inferred subsets + 4 supporting credit types.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P00

IRI và phiên bản 3.0.0

P00_ontology_header.png

1. Mở ontology/Movie_Knowledge_Graph.owl.
2. Chọn Active Ontology / Ontology Header.
3. Giữ IRI và versionInfo 3.0.0 trong ảnh.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Work and Film: meaningful inferred subsets

No unsupported feature, animation or documentary declarations

dbo:Work

dbo:Film · 30

- ActionFilm · 12
- AwardWinningFilm · 26
- AwardWinningActionFilm · 10
- GenreCrossingFilm · 8

ẢNH PROTÉGÉ CẦN BỔ SUNG · P01

Cây Work / Film

P01_film_hierarchy.png

1. Mở Classes → dbo:Work → dbo:Film.
2. Hiện ActionFilm, AwardWinningFilm và các lớp giao.
3. Giữ IRI DBpedia của Film trong Description.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Membership is inferred from source-supported facts; genre buckets are explicit author mappings.

People and companies: reuse DBpedia hierarchy

Actor and Filmmaker may overlap; roles remain individuals

dbo:Agent

dbo:Person · 851

dbo:Artist → dbo:Actor · 769

dbo:MovieDirector · 17

dbo:Writer → ScreenWriter · 34

dbo:Producer · 57

Filmmaker · 89; WriterDirector · 10

dbo:Organisation

dbo:Company · 45

ẢNH PROTÉGÉ CẦN BỔ SUNG · P02

Cây chủ thể DBpedia

P02_agent_hierarchy.png

1. Mở dbo:Agent → dbo:Person và dbo:Organisation.
2. Hiện Artist → Actor; Organisation → Company.
3. Mở MovieDirector / Writer → ScreenWriter.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

No ex:Actor / Organization / ProductionCompany clones; four role values describe credit records.

Genre, awards and provenance

Explicit label mappings are distinguished from logical entailment

dbo:Genre

dbo:MovieGenre

ex:ActionGenre · 3 genre entities

ex:DramaGenre · 9 genre entities

dbo:Award · 672 entities

Film and person award facts

No guessed award subclasses

ẢNH PROTÉGÉ CẦN BỔ SUNG · P03

MovieGenre và Award

P03_genre_award_hierarchy.png

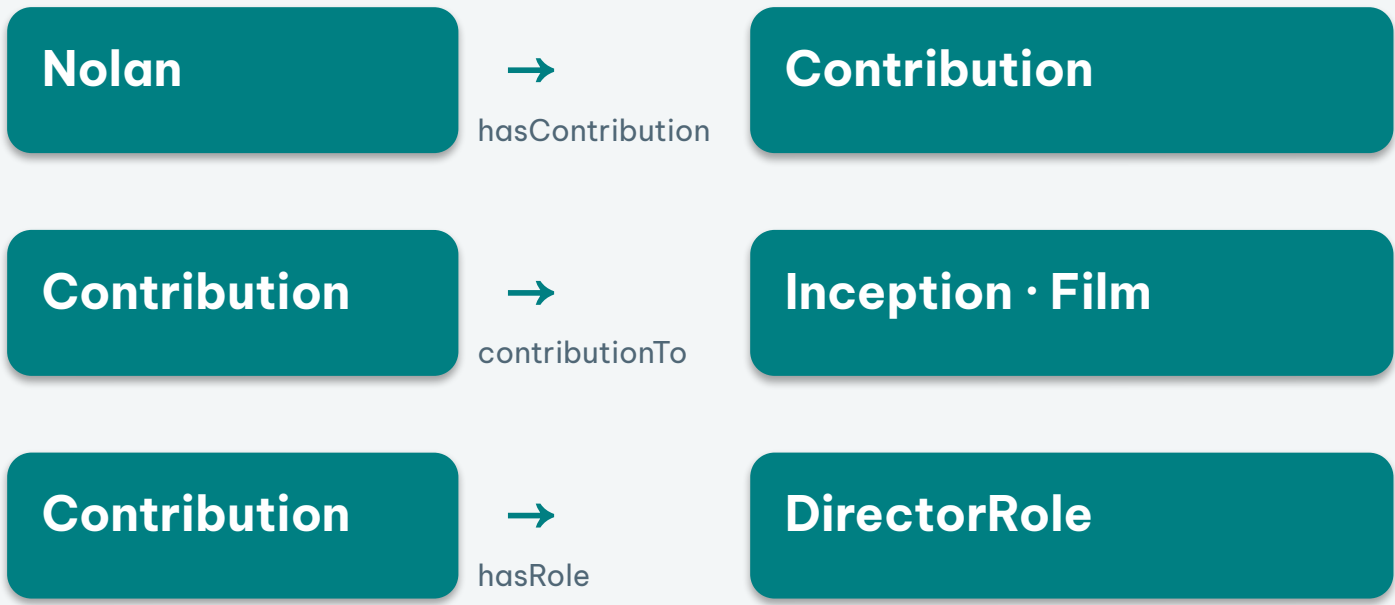
1. Mở dbo:Genre → dbo:MovieGenre.
2. Hiện ActionGenre / DramaGenre; chọn dbo:Award.
3. Giữ Description để đọc IRI và lớp cha.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Supporting vocabulary: dbo:Country / Language · ex:SourceSnapshot · void:Dataset

Contribution: person, film and role

Exactly one endpoint of each kind per credit record



Each role has its own record

Director / Actor / Writer / Producer. Nolan has 3 roles in Inception.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P04

Ràng buộc Contribution

P04_contribution_restrictions.png

1. Chọn ex:Contribution.
2. Hiện exactly 1 cho contributionBy, contributionTo, hasRole.
3. Chọn hasRole để thấy Functional và role individual.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Properties: reuse, inverse and chain

19 object properties; custom relations describe credit context

Property	Domain → range	Inverse
contributionBy	Credit → Person	hasContribution
contributionTo	Credit → Film	contributionOf
hasRole	Credit → Role	functional
dbo:director	Film → Person	ex:directed
dbo:starring	Work → Actor	ex:actedIn
contributedTo	Person → Film	property chain

ẢNH PROTÉGÉ CẦN BỔ SUNG · P05

Domain / range / inverse

P05_object_property.png

- 1. Chọn contributionBy: Contribution → dbo:Person.
- 2. Hiện Functional và inverse hasContribution.
- 3. Có thể chụp contributedTo với property chain.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

contributionBy / contributionTo / hasRole: functional. Domain/range can infer types; inverse reverses a relation.

OWL axioms: what each statement means

Model semantics and structural data validation answer different questions

Class definitions

AND / OR / SOME / VALUE
Conditions produce types not asserted in source facts.

Exactly one endpoint

Contribution: one person, film and role.
Missing-field checks are separate.

Inverse and property chain

hasContribution o contributionTo → contributedTo
965 local person–film pairs.

Open world and identity

Missing facts do not imply falsehood.
Different IRIs do not establish inequality.

Verified domain classes after HermiT

All listed types have zero assertions in the input data

Class	Condition in plain language	Count	Method
Filmmaker	Directing, writing or producing credit	89	HermiT
ActionFilm	Action genre membership	12	HermiT
AwardWinningFilm	Film with an award	26	HermiT
MultiCreditContributor	At least 2 provably distinct credits	17	HermiT
ThreeCreditContributor	At least 3 provably distinct credits	7	HermiT
WriterDirector	MovieDirector AND ScreenWriter	10	HermiT
ActorFilmmaker	Actor AND Filmmaker	7	HermiT
AwardWinningFilmmaker	Filmmaker with an award	65	HermiT
AwardWinningActionFilm	ActionFilm AND AwardWinningFilm	10	HermiT
GenreCrossingFilm	Action AND drama genre membership	8	HermiT

Supporting credit types: Acting 855 · Directing 31 · Writing 51 · Producing 73; dbo:Actor 769.

Nolan: facts → credit type → Filmmaker

A multi-step explanation grounded in real records

SOURCE FACTS

Nolan a dbo:Person; credit hasRole DirectorRole.



CREDIT CLASS

credit a DirectingContribution; inverse yields
hasContribution.



PERSON CLASS

Nolan a Filmmaker; directing + writing also yield
WriterDirector.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P06

Định nghĩa Filmmaker

P06_filmmaker_equivalent_class.png

1. Chọn ex:Filmmaker.
2. Hiện Equivalent To với Person AND các nhánh SOME.
3. Đọc các nhánh Directing / Writing / ProducingContribution.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Nolan: 8 directed films, 22 credit IRIs in the sample; three roles on Inception.

Genuine minimum-cardinality reasoning

Functional hasRole + different roles prove distinct credits

MultiCreditContributor · 17

Person and hasContribution min 2 Contribution.
Distinctness needs semantic evidence.

ThreeCreditContributor · 7

Person and hasContribution min 3 Contribution.
Nolan supplies three different-role witnesses.

Distinct IRIs ≠ provably different OWL individuals.

Negative test: min 2 genres gives 0 DL members; SQL/SPARQL counting is a different claim.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P07

Cardinality trên Contribution

P07_three_credit_cardinality.png

1. Chọn ex:ThreeCreditContributor; hiện min 3 hasContribution Contribution.
2. Chạy HermiT; chọn Nolan ở inferred view.
3. Chụp thêm Functional hasRole và AllDifferent của bốn role.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Source claims and reproducibility

Real crawl fields; no fabricated budget, ratings or awards

Source-supported relations

P57 / P161 / P58 / P162: credits.
P136 / P166 / P272: genre, award, company.

Normalized values

P2047 duration → seconds.
P577 → earliest Gregorian release-year summary.

Traceability

76 responses; retrieval URL / time / SHA-256.
Identity links and provenance have different meanings.

R2 · Actual Wikidata response and provenance

Read from the saved source file; SHA-256 verified. This is a source viewer, not a Protégé screenshot.

Response metadata

```
{
  "provider": "Wikidata",
  "url": "https://www.wikidata.org/w/api.php?action=wbgetentities&format=json&languages=en&sites=enwiki&titles=Inception%7CInterstellar+%28film%29%7CThe+Dark+Knight%7CMemento+%28film%29%7CDunkirk+%282017+film%29%7CThe+Prestige%7CBatman+Begins%7CTenet%7CThe+Godfather%7CThe+Godfather+Part+II&props=labels%7Cdescriptions%7Csitelinks%7Cclaims",
  "retrieved_at": "2026-10-06T08:43:21.455429+00:00",
  "sha256": "56d8b57c83cb1ad3dc2d2e0cf47035801fede896cc0dca296a6e8ff1e0c246e1"
,
  "path": "data/raw/1db5b6a369cc85140c7afe5e.json",
  "http_status": 200
}
```

Selected raw fields (references omitted)

```
{
  "id": "Q25188",
  "labels": {
    "en": {
      "language": "en",
      "value": "Inception"
    }
  },
  "P57_director": [
    {
      "mainsnak": {
        "snaktype": "value",
        "property": "P57",
        "hash": "c163ceb80831ed3f753fb3b90ec8c31d153eb0bf",
        "datavalue": {
          "value": {
            "entity-type": "item",
            "numeric-id": 25191,
            "id": "Q25191"
          },
          "type": "wikibase-entityid"
        },
        "datatype": "wikibase-item"
      }
    }
  ]
}
```

Right: actual Wikidata metadata and response excerpt. Hashes verify integrity, not real-world truth.

RDF literals and DBpedia units

runtime is seconds; labels are annotations

```
res:film-Q25188 a dbo:Film ;
rdfs:label "Inception"@en ;
ex:releaseYear 2010 ;
dbo:runtime "8880.0"^^xsd:double ;
dbo:director res:person-Q25191 .
```

Data property	xsd: type
dbo:runtime	double · seconds
ex:releaseYear	integer
ex:sourceUrl	anyURI
ex:retrievedAt	dateTime
ex:sha256	string

ẢNH PROTÉGÉ CẦN BỔ SUNG · P08

Thời lượng theo DBpedia

P08_runtime_datatype.png

- 1. Chọn dbo:runtime trong Data properties.
- 2. Hiện domain dbo:Work và range xsd:double.
- 3. Chọn Inception: runtime = 8880 giây.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Linked data: identity and publication

Canonical identifiers, graph exports and external identity links

1★	Open license	CC BY-SA 4.0
2★	Structured source descriptions	Claims, identifiers, values
3★	Open RDF serializations	RDF/XML, Turtle, JSON-LD
4★	HTTP identifiers and RDF	Canonical version 3.0.0
5★	External identity links	1,699 Wikidata + 28 DBpedia

Public graphs and local canonical exports are checked for semantic equality.

Inception: RDF resource and OWL

The web description and final OWL share the same source facts

← MovieLOD

Inception

<https://laminhtrung.github.io/movie-lod-semantic-web/resource/film-Q25188>

Download RDF (Turtle)

Property	Value
award	Hugo Award for Best Dramatic Presentation, Long Form
award	Saturn Award for Best Science Fiction Film
award	Academy Award for Best Cinematography
award	National Board of Review: Top Ten Films
award	Academy Award for Best Visual Effects
award	Academy Award for Best Sound Editing
award	Academy Award for Best Sound
country	United Kingdom
country	United States
director	Christopher Nolan
genre	mystery film
genre	drama film
genre	action film
genre	science fiction action film
genre	thriller film
genre	adventure film

ẢNH PROTÉGÉ CẦN BỔ SUNG · P09

Cá thể Inception bản mới

P09_inception_individual.png

- 1. Mở Movie_Knowledge_Graph.owl 3.0.0.
- 2. Chọn film-Q25188; hiện label, releaseYear 2010, runtime 8880.
- 3. Hiện dbo:director, genre, productionCompany và award.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Left: current version-3 web description. Right: capture the same individual in Protégé.

SPARQL: choose the knowledge scope

Source facts, OWL inference or before/after named graphs

```
SELECT ?year ?seconds ?director WHERE {
  res:film-Q25188 ex:releaseYear ?year ;
  dbo:runtime ?seconds ; dbo:director ?person .
  ?person rdfs:label ?director .
}
```

2010 • 8,880 seconds • Nolan

SELECT chooses columns; WHERE matches; OPTIONAL preserves rows.

The sample chooses its required scope automatically; change scope to compare before and after.

MovieLOD

QueriesFilmsOntologyDataset

Download RDF ↗

SEMANTIC WEB / MOVIES

Ask questions about movie data.

301,72737

filmsexternal linksclasses

Explore ontology 3.0.0: source facts, inferred knowledge and links to Wikidata and DBpedia.

SPARQLQuery language

Sample queries

Inception: year, duration and director

Knowledge scope

Source facts

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>

PREFIX dbo: <http://dbpedia.org/ontology/>

PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?title ?year ?runtimeSeconds ?director WHERE { res:film-Q25188 rdfs:label ?title ; ex:releaseYear ?year ; dbo:runtime ?runtimeSeconds ; dbo:director ?person . ?person rdfs:label ?director }

Run queryCtrl ⌘

Copy

SELECT: get a table · ASK: test a condition · CONSTRUCT: create an RDF graph.

Results

Download results

1 result · 0.12 s

title	year	runtimeSeconds	director
Inception	2010	8880	Christopher Nolan

Querying the RDF dataset in your browser with Comunica.

Reading an RDF statement: Inception → director → Christopher Nolan. ex: is the project vocabulary; dbo: is the DBpedia vocabulary; variables start with ?.

Credit records and competency questions

27 design queries; counts use local identities

MovieLOD

QueriesFilmsOntologyDataset

Download RDF ↗

SEMANTIC WEB / MOVIES

Ask questions about movie data.

Explore ontology 3.0.0: source facts, inferred knowledge and links to Wikidata and DBpedia.

30

1,727

37

filmsexternal linksclasses

SPARQL

Query language

Sample queries

Inception: year, duration and director

Knowledge scope

Source facts

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#> PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/> PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> SELECT ?credit ?role WHERE { res:person-Q25191 ex:hasContribution ?credit . ?credit ex:contributionTo res:film-Q25188 ; ex:hasRole ?r . ?r rdfs:label ?role } ORDER BY ?role

Run query ⌘+R

Copy

SELECT: get a table · ASK: test a condition · CONSTRUCT: create an RDF graph.

Results

Download results

3 results · 0.06 s

credit	role
contribution-Q25188-Q25191-director	Director
contribution-Q25188-Q25191-producer	Producer
contribution-Q25188-Q25191-writer	Writer

Querying the RDF dataset in your browser with Comunica.

Reading an RDF statement:

Inception → director → Christopher Nolan.

ex: is the project vocabulary; dbo: is the DBpedia vocabulary; variables start with ?.

25 Inception credits

21 acting + 1 directing + 1 writing + 2 producing.

8 / 4 / 7

Nolan films / Inception companies / Godfather awards.

27 design queries

Direct facts / hierarchy / inferred knowledge.

Data checked: 09 Oct 2026 • Sources: project code and evidence/

20 / 24

Before / after: Nolan and WriterDirector

The same result is available in web, local endpoint and terminal

```
.venv/bin/python src/query.py queries/20.rq
.venv/bin/python src/query.py queries/20.rq --reasoned
.venv/bin/python src/query.py queries/27.rq --mode dataset
```

Mode	Nolan types returned
Asserted	WriterDirector 0; Nolan min 3: false
Reasoned	WriterDirector 10; Nolan min 3: true

ẢNH PROTÉGÉ CẦN BỔ SUNG · P10

Các type suy luận của Nolan

P10_nolan_inferred_types.png

- 1. Chọn HermiT → Start reasoner, chờ hoàn tất.
- 2. Chọn person-Q25191; mở inferred types.
- 3. Giữ WriterDirector, Filmmaker, ThreeCreditContributor trong ảnh.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

All query interfaces read the same canonical graph exports; inference is computed before serving.

Verification of the final OWL

Direct HermiT run; evidence bound to the file hash

PASS

HermiT consistency

0

unsatisfiable named classes

27

design queries executed

965

chain pairs verified

Final OWL: HermiT verification

Direct RDF/XML input · version 3.0.0 · counts restricted to local IRIs

Consistency	PASS
Unsatisfiable named classes	0
Design SPARQL queries	27
Property-chain person-film pairs	965
MultiCredit / ThreeCredit	17 / 7
WriterDirector	10
Web publication scope	Canonical 3.0.0

Author-rendered evidence summary; not a screenshot of Protégé.

Canonical OWL verified by HermiT; browser queries use matching source, reasoned and named graphs.

Coverage and limitations

Verified functional requirements; evidence scope remains explicit

Req.	Evidence	Status
R1	Canonical OWL + HermiT	Verified
R2	Source facts + provenance	Verified
R3	Public RDF / OWL	Aligned
R4	Retained identity links	Verified
R5	Three query scopes	Verified

Canonical model and application are aligned.

Inference scope

Consistency does not prove completeness or factual accuracy.

Source and identity

No blanket inequality for QIDs; genre buckets are author mappings.

Deliverables

Slides, report and guides are current. No MP4 files are included.

Evidence capture and submission checklist

Report: 15 pages · main deck: 24 slides · no MP4 included

Image	Slide	Evidence to capture
P00	5	Ontology 3.0.0 header
P01	6	Work and Film hierarchy
P02	7	DBpedia agent hierarchy
P03	8	MovieGenre and Award
P04	9	Contribution restrictions
P05	10	Inverse and functional properties
P06	13	Filmmaker equivalent class
P07	14	Three-credit cardinality
P08	16	DBpedia runtime in seconds
P09	18	Inception in the final OWL
P10	21	Nolan inferred types

Before submission

1. Verify ontology version 3.0.0.
2. Insert the requested genuine screenshots.
3. Save them in evidence/protege/ for automatic replacement.
4. Rehearse the script and demonstrate the current query modes.

Use the canonical OWL for Protégé evidence; source facts and inferred knowledge have separate scopes.